

Multi-Agent Workflow Design – Pharmacy Agent Assistant

System Overview

The Pharmacy Agent Assistant is designed to support pharmacy technicians during the prescription pickup process by providing quick access to relevant patient information and highlighting important details that may require attention. In fast-paced retail pharmacy environments, technicians are often responsible for handling multiple tasks simultaneously, which increases the likelihood of overlooking critical information such as medication issues, insurance delays, or prior complications. To address this challenge, the system has been extended into a multi-agent AI architecture that distributes responsibilities across specialized agents, each performing a specific function within the workflow.

Rather than relying on a single system to process all tasks, the multi-agent approach introduces modularity and improves reliability by allowing each agent to focus on a well-defined role. The system retrieves patient data, analyzes potential risks, generates a clear summary, and verifies the output before presenting it to the user. This layered structure ensures that the final information displayed to the pharmacy technician is both accurate and useful for decision-making during prescription pickup.

Agent Architecture and Workflow

The system follows a structured pipeline architecture coordinated by a central controller, which manages the flow of information between agents. When a user enters a customer's name into the front-end interface, the controller initiates a sequence of operations that pass data through each agent in a fixed order. This pipeline consists of four main stages: data retrieval, risk analysis, summary generation, and safety validation.

The process begins with the retrieval of the customer's profile, which is then analyzed to identify any relevant risks or issues. These identified risks are used to generate a concise summary tailored to the needs of the pharmacy technician. Finally, the system validates the generated summary to ensure that no critical information has been omitted. If any issues are detected during validation, the system has the ability to regenerate the summary, creating a limited feedback loop that improves output quality.

Agent Roles and Responsibilities

Each agent in the system is designed to perform a distinct and specialized role. The Data Retrieval Agent acts as the entry point into the system's data layer, receiving the customer's name as input and returning a structured profile containing demographic information, medications, past issues, insurance details, immunizations, and diagnoses. This agent ensures that all relevant data is gathered before further processing begins.

Once the customer profile is retrieved, it is passed to the Risk Analysis Agent, which serves as the domain expert within the system. This agent examines the patient's data to identify potential concerns that may impact the prescription pickup process. For example, it detects patterns such as missed refills, insurance delays, or cases involving multiple medications that may require additional review. The output of this stage is a set of clearly defined risk flags that highlight important issues for the technician.

The Summary Generation Agent then takes both the customer profile and the identified risks and transforms this structured information into a concise, natural language summary. This agent plays a critical role in making the system user-friendly, as it translates complex data into a format that can be quickly understood in a busy work environment. The generated summary emphasizes key concerns while still providing an overview of the patient's medication status.

The final stage of the workflow is handled by the Safety and Compliance Agent, which functions as a critic and verifier. This agent reviews the generated summary and checks whether all identified risks have been properly included. Its primary goal is to ensure that the output meets safety standards and does not omit critical information that could impact patient care. If the summary is deemed incomplete or inaccurate, the system can trigger a revision process to regenerate the output.

Communication and Data Flow

Communication between agents occurs through structured data formats that allow each stage of the pipeline to operate independently while still contributing to the overall workflow. The Data Retrieval Agent outputs a customer profile in the form of a structured data object, which is then passed to the Risk Analysis Agent. The Risk Analysis Agent produces a list of risk flags that are forwarded to the Summary Generation Agent along with the original profile data. The Summary Agent outputs a text-based summary, which is then evaluated by the Safety Agent.

This clear separation of inputs and outputs ensures that each agent can be modified or improved without affecting the entire system. It also allows for greater transparency, as each stage of processing can be independently examined and validated.

Failure Handling and Quality Control

To ensure robustness, the system includes several mechanisms for handling errors and low-quality outputs. If a customer's name does not match any records in the dataset, the system returns a clear error message to the user. In cases where data may be incomplete or missing, the system defaults to safe and conservative outputs to avoid presenting misleading information.

The Safety and Compliance Agent plays a key role in quality control by verifying that all relevant risks are included in the summary. If discrepancies are detected, the system initiates a regeneration step to improve the output. Additionally, if any agent fails to produce a valid result, the controller can provide a fallback response to maintain system functionality. These safeguards help ensure that the system remains reliable even in edge cases.

Example Workflow

To illustrate the system in action, consider a scenario where a pharmacy technician enters the name "John Doe" into the interface. The Data Retrieval Agent first locates the corresponding profile and returns the patient's information. The Risk Analysis Agent then evaluates the profile and identifies issues such as a history of missed refills and insurance delays. The Summary Generation Agent uses this information to create a concise pickup summary that highlights these concerns. Finally, the Safety and Compliance Agent reviews the summary to ensure that all risks are included before the final output is displayed to the technician.

Conclusion

The multi-agent architecture significantly enhances the functionality of the Pharmacy Agent Assistant by introducing specialization, structure, and validation into the system. By dividing responsibilities across multiple agents, the system becomes more reliable, scalable, and easier to maintain. Each agent contributes a specific layer of intelligence, resulting in a more effective tool for supporting pharmacy technicians in high-pressure environments. This approach not only improves efficiency but also helps reduce the risk of errors, ultimately contributing to safer and more informed patient interactions.